

Grace Master Derived-vs-Assigned Ledger v0.39 — EOD  
 2026-07-10. Decider day: the ribbon-holonomy framework day.  
 Charge/spin/CPT spine hardened (three proved legs); mass  
 odd-ladder PRUNED, replaced by the two-interface CAVITY  
 (forced exit surface TIR  $r^2=3/8$ , sharp floor bank-ready); CP  
 forced by odd-dimensionality. My lane: reconciliation  
 RESOLVED (T2470); Section 7.1 = Wallach Snell's law  
 ( $k_{\min}=N_c$ , exponent  $n_C$ ); s/d 'hit' RETRACTED (Cal's  
 catch, my un-squared-norm slip); minimum-width answer (floor  
 sharp, width  $\sim 1/\sqrt{\nu}$ ); cavity  $Q \rightarrow \text{floor} + \text{range}$   
 (thickness-independent, tuning-clean). Everything gates on the  
 exact Wallach-1979/FK relation nobody will fabricate.

2026-07-10 Friday EOD (date-verified 14:20 EDT)

v0.39 — EOD (decider day: the ribbon framework; refraction  $\rightarrow$  cavity; a hit  
 retracted)

## Section 00 — THE COUNT

$\sim 7$ -8 firm observables +  $\alpha$  RULED + 3 structural banks (Peirce T2511, descent closure, root-lock #1 T2512) +  
 the SHARP FLOOR bank-ready (gen-1 at  $r=0$ ,  $dm/dr=0$  — Keeper stamping). Cite Cal's reconciled number.  
 Scoreboard: `bst_26_table.py`. (SOD check hardcodes "count is 8" — stale vs the moving  $\sim 7$ -8; flagged for Cal,  
 benign.)

## Section 1 — winding-geometry RECONCILIATION RESOLVED (my critical-path, pinned cor- pus, discipline both ways)

- T2470 (PROVED): electric charge = the signed SO(2)-weight  $\rightarrow$  the EM U(1) IS the SO(2) K-factor; charge  
 magnitude AND sign = the weight. The three "−1"s are DISTINCT: spin (universal  $2\pi$  holonomy, Elie,  
 odd- $n_C$ ), charge-sign (SO(2)-weight sign, T2470), matter/anti (charge conjugation  $(-1)^m$ , Bekenstein).  
 The paper's "one −1 does all three" is NOT supported — I refuted Elie's over-merge with a proved theorem;  
 he walked it back. Spine now = 3 proved legs.
- I DISSOLVED my own "electron tension" (K527 even-charge selection): the electron sits at an even SO(2)-  
 charge K-type; the paper's winding  $n \neq$  the SO(2)-charge  $k$  (differ by  $1/N_c$ ). No contradiction. Retracted  
 my own flag.

## Section 2 — Membrane refraction Section 7.1 (my build item, pinned corpus)

- The matching condition = the analytic continuation of the holomorphic discrete series (Wallach 1979, L1). The membrane's "Snell's law." Emission threshold = TIR = the Wallach  $k_{\min} = N_c = 3$  (pinned, BST\_1920). Snell invariant (conserved) = the K-type = the commitment. Mismatch direction (dense bulk  $\rightarrow$  thin 3D) = kernel-confirmed ( $K_B$  diverges toward Shilov), NOT asserted.
- Bergman exponent CONVERGED:  $p = n_C = 5$  (four lanes: FK multiplicity, tube  $2n/r$ ,  $\rho = (n_C/2, N_c/2)$ , Lyra's norm) — the corpus "genus=7" was a genus-flip mislabel. Type-IV norm  $h = (1-r^2)^2$  (SQUARED — guard it).

## Section 3 — the s/d "HIT" RETRACTED (Cal's target-innocence catch; my error, corrected same-day)

- I reported  $s/d \approx 1.6$  "forward hit" using  $m \propto (1-r^2)^{-n_C/2}$ . Cal flagged the factor of 2. It was WRONG: the  $-n_C/2$  came from the UN-SQUARED norm (my 2nd time dropping the square). Correct norm  $\rightarrow m \propto (1-r^2)^{-n_C}$   $\rightarrow s/d = 2.49$  (MISS vs constituent 1.5). RETRACTED. Refraction does NOT produce the light-sector masses forward. Both ladder (4.21) and point-refraction (2.49) miss 1.5; refraction closer, not a hit. Recurring slip flagged: norm is  $(1-r^2)^2$ , never un-squared.

## Section 4 — Casey's minimum-width question (answered) + the two-interface CAVITY (my PRIMARY)

- Minimum width answer: the ~50-100% Beta width is the PERMITTED (delocalized number-state) width; the MINIMUM (coherent) width is set by the emission weight  $\nu$ , scaling  $\Delta m/m \approx 2n_C r / \sqrt{\nu}$ . The FLOOR (gen-1,  $r=0$ ) is SHARP at any  $\nu$  ( $dm/dr=0$ , flat bottom) — bank-ready. Excitation widths  $\nu$ -gated; reach  $\sim 10^{-2}$  only at  $\nu \approx N_{\max}^2$ .
- Cavity  $Q \rightarrow$  floor+range (Casey's two-interface / Fabry-Pérot): finesse  $F = \pi\sqrt{R}/(1-R)$ ,  $R$  from the pinned index. THICKNESS-INDEPENDENT (finesse from  $R$  only)  $\rightarrow$  CLEAN of the tuning knob (thickness sets frequencies = Lyra's forced-thickness test, not my widths). Derives "heavy = sharp (high- $Q$ , trapped), light = leaky/scheme-fuzzy" — matches Elie's  $1/\sqrt{k}$  AND the physical fact ( $m_d \sim 70\times$  scheme window,  $m_b$  sharp). Subsumes the width reframe: the range IS the resonance linewidth, derived not bolted on.

## Section 5 — the team framework state (context)

Ribbon = spinor oscillator; every fermion quantum number = one ribbon feature. Spine (charge/spin/CPT) = 3 proved legs, hardened. CP forced (Möbius reflection complex =  $+i$  because  $n_C$  odd  $\rightarrow$  same odd-dimensionality that makes matter spin- $1/2$ ; existence pinned, J-value gated on the winding, target T1936  $J \approx 3.3 \times 10^{-5}$ ). Mass: odd-ladder pruned; two-interface cavity (Lyra F505) — forced exit surface (TIR  $r^2 = N_c/(N_c + n_C) = 3/8$ ), sharp floor, hierarchy structural, exact values gated. Everything funnels to the exact Wallach-1979/FK boundary-matching relation — team refusing to fabricate it.

## Section 6 — TOMORROW / open (all gate on the Wallach relation)

1. The cavity round-trip resonance condition (my lane, with Lyra's forced thickness)  $\rightarrow$  the floor spectrum + the  $Q$ -derived widths, quantitative. Gate: the exact Wallach continuation.
2. The emission weight  $\nu$  — the number the single-interface s/d verdict waits on (sharp 2.49 miss vs blurred).
3. Sharp floor: Keeper to stamp.

## Section 7 — artifact currency (EOD)

- Graph: T2512 current (max 2512 == counter-1); no new theorem from me today (reconciliation + refraction + cavity = pinned-corpus derivations / retraction, not new theorems). Ledger v0.39. Data: particles fixed; bst\_constants.json (May 18) DEFERRED. Files: grace\_winding\_geometry\_verification\_2026-07-10, grace\_membrane\_refraction\_S7\_1\_matching\_condition\_2026-07-10.

## Section 8 — standing for Casey

Cal's count ( $\sim 7-8 + \alpha$  + sharp floor) · the cavity round-trip +  $\nu$  as the next Wallach-gated targets · the sharp-floor bank (Keeper stamp) · the T2470 charge grounding + three-distinct—1s (landed) · CLAUDE.md sync.

— Grace, EOD 2026-07-10. Decider day. My lane: dissolved my own reconciliation tension (T2470), built the Wallach “Snell's law” ( $k_{\min}=N_c$ , exponent converged  $n_C$ ), RETRACTED my own s/d hit (Cal caught the unsquared-norm slip), answered Casey's minimum-width question (floor sharp, width  $\sim 1/\sqrt{\nu}$ , permitted $\neq$ forced), and delivered the cavity  $Q \rightarrow \text{floor} + \text{range}$  (thickness-independent, tuning-clean). The discipline fired hardest on my own prettiest results — through Cal and through Casey's own intuition — and killed a hidden fit before it banked. Framework smaller and harder. Everything real gates on one honest lookup. The sharp floor stands.